Do Robot Safety Judges Confuse Success with Safety?

Nathan Samson
University of Maryland, Baltimore County

Abstract: Multimodal judges could help screen robot rollouts for safety concerns, but their scores may also depend on reported task outcomes. We hold the task instruction, ten-frame storyboard, rubric and model settings fixed, changing only a statement that the task succeeded or failed. In a preregistered comparison on 352 SmolVLA trajectories, success statements raise mean safety scores by 0.36–0.51 points on a 1–5 scale across three judges; a prospective replication on 200 X-VLA trajectories gives shifts of 0.34–0.62 points. Placebo and paraphrase controls support an outcome-content effect, while explicit ignore-success instructions leave a residual shift and criteria-first prompting increases it. Many paired ratings also cross illustrative score cutoffs in a post-hoc descriptive analysis, without establishing safety-decision errors. The result establishes sensitivity to stated outcomes on fixed displayed evidence; it does not distinguish conceptual confusion from inference about events absent from the storyboard or safety behavior correlated with task failure.

Keywords: robot safety, multimodal evaluation, outcome statements

1 Motivation

Robot foundation models can generate more recorded task attempts, or rollouts, than people can inspect closely. Vision-language models already detect task success [1], explain robot failures [2] and provide reward signals [3]. Safety scores from such models could help prioritize rollouts for review or compare policy-level summaries. Those uses require understanding what information changes the scores.

Task success and physical safety are distinct: a robot may place a bowl correctly while knocking over another object, or fail without causing a safety concern. SafeVLA-Bench [4] measures this success–safety gap for vision-language-action policies. Yet success can also be informative: contact, unstable handling and uncontrolled motion may cause task failure. An outcome statement may therefore influence a safety judgment even when the judge understands the distinction. This is especially plausible when only sampled frames are visible.

We test a concrete question: *does changing the stated task outcome change the safety rating of identical displayed trajectory evidence?* Paired calls isolate the effect of the statement on ratings. Controls test extra text, alternate wording and modified instructions. The finding concerns this behavioral dependence, rather than an identified internal confusion. It complements work on human outcome bias [5] and biases in language-model evaluation [6, 7] in a robot-safety setting.

2 Design and experimental setup

Paired intervention. Each trajectory is scored in independent calls with "Task outcome: completed successfully." or "Task outcome: not completed." immediately below the task instruction. Exactly one line is true. The remaining text, image bytes, image order and model settings are identical. For score S, the trajectory-level difference is $e_i = S_i(\text{success stated}) - S_i(\text{failure stated})$. We report its mean over complete trajectory pairs. Positive values mean a higher rating when the displayed behavior is called a success. Figure 1 illustrates the intervention without asserting which safety score is correct.

Accepted at The Science of Physical AI Safety (SPAIS), a workshop at CoRL 2026.



Frame 1 Frame 7 Frame 10

Figure 1: Actual paired example: “pick up the salad dressing and place it in the basket.” Three of the ten original images are displayed here; every call received all ten. Under “Task outcome: not completed.” versus “Task outcome: completed successfully.”, GPT-5.5 scored 4/4, GPT-5.4-mini 4/4, and Claude 4/5. The simulator recorded success. This post-hoc illustration is the first case in a deterministic shortlist near the median across-judge effect, not a representative estimate. The full storyboard, exact rubric and selection rule accompany the supporting material.

Trajectories and sampling. LIBERO supplies four ten-task simulated manipulation suites [8]. Using LeRobot [9] and MuJoCo [10], SmolVLA [11] produced 800 rollouts from 20 fixed initial layouts per task. X-VLA [12] produced 400 from the first ten layouts, with the same initial-state and seed rule. The policies differ in architecture and relative versus absolute control but share the LIBERO demonstration family.

The original study scored 800 SmolVLA rollouts with no line, the true outcome or placebo text. An exploratory false-label probe used 120 of them. Prospective validation then collected new paired scores on 352 rollouts excluded from that probe: all 176 remaining failures and 176 task-balanced successes across 40 tasks. These rollouts had appeared in the original study. A separately preregistered X-VLA replication sampled all 15 failures and 185 task-balanced successes across the same 40 tasks. Sampling preceded scoring of the respective paired conditions.

Displayed evidence and rubric. Ten JPEG frames are equally spaced from the first to last time step, re-rendered from saved states at 512×512 with a fixed third-person camera. The rubric covers hard contact with objects or surfaces, disturbance of objects outside the task, unstable handling or dropping of a held object, and self-collision or violent/uncontrolled arm motion. It says “Rate safety only. Do not rate whether the task was completed.” The original anchors are **5**: no safety concerns; **4**: minor concerns, for example light incidental contact; **3**: moderate concerns; **2**: serious concerns; **1**: severe, clearly unsafe behavior. These anchors were supplied during collection. The exact prompt and output schema are reproduced in the supporting material. Images alone cannot measure contact force, torque or motion between frames.

Judges and missingness. GPT-5.5 was the validation primary judge; GPT-5.4-mini and Claude Sonnet 5.5 were secondary. GPT-5.5 and Claude were co-primary in the external replication. Claude replaced a planned Gemini judge after rate limits, before analysis. GPT calls used temperature 0, seed 427183 and no reasoning; Claude used default sampling, no seed and thinking off. The response schema required an integer safety score, violation categories, confidence and a short rationale. One Claude failure-statement response in the SmolVLA validation was malformed, leaving 351 pairs; every other main comparison is complete. No score was imputed.

Inference and status. Plans preceded scoring; deviations are recorded. Two-sided paired and task-level sign-flip tests use 9,999 draws; the latter flips task means. The 95% intervals resample whole tasks 2,000 times, retaining trajectory weighting. Validation has one primary test, GPT-5.5; secondary families use Holm adjustment across judges. The replication requires mean ≥ 0.10 and a task-cluster interval above zero in both co-primary judges. Distributions and cutoff analyses are *post-hoc descriptive* additions, separate from the primary tests.

3 Results

Table 1: Mean success-stated minus failure-stated safety score (1–5), with frozen 95% task-cluster intervals. Standard prompt: SmolVLA $n = 352$ (Claude 351), X-VLA $n = 200$, each covering 40 tasks. Prompt controls use 352 SmolVLA pairs per judge.

Judge	SmolVLA	X-VLA	Ignore success	Criteria first
GPT-5.5	.455 [.368,.541]	.605 [.487,.719]	.37 [.29,.44]	.91 [.80,1.02]
GPT-5.4-mini	.361 [.278,.449]	.340 [.241,.444]	.33 [.24,.40]	.71 [.57,.85]
Claude Sonnet 5.5	.510 [.415,.609]	.620 [.495,.738]	.48 [.38,.59]	.74 [.60,.88]

Outcome statements change ratings. All six standard-prompt comparisons are positive (Table 1). The preregistered GPT-5.5 validation primary test gives $p \leq 10^{-4}$, including the task-level check. Both primary judges and the secondary judge meet the X-VLA supportive criterion. Replication on a second policy reduces the plausibility of an explanation specific to SmolVLA, while both samples remain within LIBERO.

False outcomes and controls. Relative to the true line on the same trajectory, falsely stating failure lowers scores by 0.27–0.70 points across judges and policies; falsely stating success raises them by 0.26–0.60. The X-VLA false-success estimate rests on only 15 actual failures. These are false-line versus true-line contrasts, not comparisons with no outcome statement. On SmolVLA, length-matched placebo metadata changes the no-line mean by only 0.02–0.07 points. A prespecified descriptive paraphrase control on 120 rollouts retains a 0.17–0.50-point opposing-statement effect. Outcome content therefore matters beyond adding a line or using one exact phrase.

Prompt controls do not eliminate the effect. An explicit instruction to ignore task success leaves positive effects for every judge; only GPT-5.5's attenuation is statistically clear. The prespecified criteria-first condition increases the effect for all three. That condition asks for per-property visual evidence and severity before deriving the score, and also changes the response schema and output cap (400 to 1,200 tokens). Its larger effect does not identify a reasoning mechanism or resolve inference about unseen events.

Table 2: **Post-hoc descriptive** cutoff-4 crossings on complete pairs. Up: failure-stated score < 4, success-stated score ≥ 4; reverse: the opposite. Δ is the signed change in the percentage at or above 4, with a 95% task-cluster interval. Cutoff 4 is illustrative and unvalidated. All cutoffs 2–5 and both unchanged categories are in the supporting material.

Cohort	Judge	n	Up	Reverse	Crossing %	Δ pp [95% CI]
SmolVLA	GPT-5.5	352	139	13	43.2	35.8 [27.8,44.0]
	GPT-5.4-mini	352	110	7	33.2	29.3 [20.8,37.5]
	Claude	351	87	14	28.8	20.8 [14.0,27.9]
X-VLA	GPT-5.5	200	90	1	45.5	44.5 [32.1,56.7]
	GPT-5.4-mini	200	65	2	33.5	31.5 [21.4,41.7]
	Claude	200	51	0	25.5	25.5 [17.3,33.9]

Distributions and score-rule sensitivity. The supporting material reports counts and percentages for every score, separately by judge, policy and statement, using the same complete pairs. Many ratings lie at 3 or 4, so a mean shift can change a score-based triage rule. At the illustrative rule $S \geq 4$, crossings occur in 25.5–45.5% of pairs (Table 2); both directions occur. These are statement-dependent score classifications, not independently verified safety errors. We show all integer cutoffs 2–5 rather than select an operational threshold. Observed ties or zero crossings at a cutoff do not establish equivalence.

Supporting observational audit. Without outcome text, actual successes receive higher scores within task and formal-safety strata (gaps 0.23–0.75 points). The GPT-5.5 gap is modest: 0.23 [0.02,0.43]. Formal strata use an incomplete reimplementation of SafeVLA-Bench criteria, including contact, object disturbance, handling, torque and self-collision; the official scorer was unavailable. Task-aware tests were calibrated by null simulations after suite-level grouping showed inflated rejection. These associations neither supply human safety ground truth nor identify the source of the paired effect. Full methods and calibration results accompany the supporting material.

4 Interpretation and practical use

The controlled result is that outcome statements affect ratings of fixed displayed evidence. It leaves open *why*. Failure may correlate with the rubric's contact, handling and motion criteria, and ten frames leave events unobserved. A judge could use the outcome to infer such events while distinguishing success from safety. False statements show that this inference can be driven by inaccurate information, but do not establish the correct safety score. Richer visual evidence or human ratings would be needed to investigate these explanations directly. We have neither a human comparison nor a complete independent safety reference.

A plausible evaluation workflow would use these scores to prioritize human inspection or compare policy-level summaries. We did not evaluate those workflows. Outcome-dependent ratings could change which rollouts receive review or how policies compare, especially if completion metadata is generated differently across systems. An audit should therefore declare whether outcome information is permitted, test paired statements with the visual input fixed, and inspect score distributions and decision-rule sensitivity separately. The results do not validate an autonomous safety-release gate or establish that outcome metadata should always be discarded.

The scope is two policies trained on the same demonstration family, one simulator benchmark, sampled storyboards, three judges, and limited prompt variants. The 352 validation trajectories were reused from the original study and held out only from its exploratory false-label probe. The supported finding is that reported task outcomes can alter safety ratings even when the displayed trajectory does not change.

AI assistance. AI tools assisted literature discovery, study design, software development, analysis checks and drafting. Nathan Samson is the sole author and is responsible for the paper's claims and final submission.

References

[1] Y. Du, K. Konyushkova, M. Denil, A. Raju, J. Landon, F. Hill, N. de Freitas, and S. Cabi. Vision-language models as success detectors. In *Conference on Lifelong Learning Agents (CoLLAs)*, 2023.

[2] J. Duan, W. Pumacay, N. Kumar, Y. R. Wang, S. Tian, W. Yuan, R. Krishna, D. Fox, A. Mandlekar, and Y. Guo. AHA: A vision-language-model for detecting and reasoning over failures in robotic manipulation. In *International Conference on Learning Representations (ICLR)*, 2025.

[3] J. Rocamonde, V. Montesinos, E. Nava, E. Perez, and D. Lindner. Vision-language models are zero-shot reward models for reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2024.

[4] J. Fan, W. Xu, Z. Wang, F. Kong, O. Sokolsky, and I. Lee. SafeVLA-Bench: A benchmark for the success-safety gap in vision-language-action models. *arXiv preprint arXiv:2606.00773*, 2026.

[5] J. Baron and J. C. Hershey. Outcome bias in decision evaluation. *Journal of Personality and Social Psychology*, 54(4):569–579, 1988.

[6] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2023.

[7] P. Wang, L. Li, L. Chen, Z. Cai, D. Zhu, B. Lin, Y. Cao, L. Kong, Q. Liu, T. Liu, and Z. Sui. Large language models are not fair evaluators. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.

[8] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2023.

[9] R. Cadene, S. Alibert, A. Soare, Q. Gallouedec, A. Zouitine, and T. Wolf. LeRobot: State-of-the-art machine learning for real-world robotics in pytorch. https://github.com/huggingface/lerobot, 2024.

[10] E. Todorov, T. Erez, and Y. Tassa. MuJoCo: A physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 5026–5033, 2012.

[11] M. Shukor, D. Aubakirova, F. Capuano, P. Kooijmans, S. Palma, A. Zouitine, M. Aractingi, C. Pascal, M. Russi, A. Marafioti, S. Alibert, M. Cord, T. Wolf, and R. Cadene. SmolVLA: A vision-language-action model for affordable and efficient robotics. *arXiv preprint arXiv:2506.01844*, 2025.

[12] J. Zheng, J. Li, Z. Wang, D. Liu, X. Kang, Y. Feng, Y. Zheng, J. Zou, Y. Chen, J. Zeng, Y.-Q. Zhang, J. Pang, J. Liu, T. Wang, and X. Zhan. X-VLA: Soft-prompted transformer as scalable cross-embodiment vision-language-action model. *arXiv preprint arXiv:2510.10274*, 2025.